

# Finite Volume Method

Scalar conservation laws used to model the evolution of a conserved quantity over time have the form

$$u_t + f(u)_x = 0. \tag{1}$$

Here,  $u(x, t)$  is the conserved scalar quantity, and  $f(u)$  denotes the flux function associated with the conserved quantity. For smooth initial data with non-intersecting characteristic curves, the method of characteristics yields an exact solution of equation (1). However, when characteristic curves intersect, shock waves form, and a classical solution does not exist. In such circumstances, it becomes necessary to introduce weak solutions, formulated through an integral representation of the conservation law. Nevertheless, weak solutions are not inherently unique, and without the enforcement of suitable admissibility or entropy conditions, they may correspond to non-physical or mathematically irrelevant solutions.

Numerical approximation methods play a crucial role in this scenario. Standard finite difference schemes often produce oscillations near shocks or select the wrong weak solution entirely. Similarly, the finite element method and spectral method cannot be used, as their approximation spaces assume smoothness. Finite volume methods, on the other hand, are naturally adapted to hyperbolic problems. In this talk, we explore how finite volume methods – particularly the upwind scheme – can be used to compute the numerical solution of hyperbolic partial differential equations such as scalar conservation laws.